

WAGGA WAGGA, Australia

WMO: 949100

Lat: 35.1583S

Lon: 147.4572E

Elev: 699

StdP: 14.33

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.3	33.4	19.7	15.2	79.6	24.7	19.2	72.4	22.5	49.5	20.1	51.4	2.9	110	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.1	100.2	65.4	96.2	64.7	92.6	64.0	70.6	84.7	69.1	83.7	67.8	82.9	11.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	104.2	73.4	65.2	95.6	72.5	63.0	88.4	72.3	34.9	84.6	33.6	83.3	32.5	83.1	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.5	17.3		27.2	105.9	1.5	3.9	26.1	108.7	25.2	111.0	24.3	113.2	23.2	116.0
				26.9	73.1	1.5	1.9	25.8	74.5	24.9	75.7	24.0	76.8	22.9	78.2
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	77.0	75.2	69.3	61.2	53.3	48.5	46.8	48.2	53.2	59.6	66.7	72.0	(6)
(7)		DBStd	12.12	7.26	6.77	6.49	6.05	5.33	4.40	3.99	4.38	5.32	6.78	7.52	7.29	(7)
(8)		HDD50	335	0	0	0	1	26	79	116	87	23	3	0	0	(8)
(9)		HDD65	2764	3	6	32	143	363	495	565	519	356	197	67	18	(9)
(10)		CDD50	4289	836	706	597	337	129	34	15	33	118	301	502	681	(10)
(11)		CDD65	1243	375	292	164	29	1	0	0	1	29	118	234		(11)
(12)		CDH74	15095	4721	3244	1703	314	6	0	0	0	39	481	1540	3047	(12)
(13)		CDH80	7208	2583	1590	659	63	0	0	0	7	152	668	1486		(13)
(14)	Wind	WSAvg	7.5	9.3	8.6	7.7	6.5	6.1	6.3	6.7	7.5	8.0	8.8	9.1	(14)	
(15)	Precipitation	PrecAvg	22.7	1.7	1.5	1.8	1.6	2.0	1.9	2.2	2.1	2.0	2.1	1.9	1.9	(15)
(16)		PrecMax	39.6	6.9	7.4	8.2	8.5	7.2	5.5	5.1	4.0	6.6	7.2	5.9	8.4	(16)
(17)		PrecMin	9.6	0.0	0.0	0.0	0.0	0.2	0.0	0.1	0.2	0.2	0.1	0.0	0.0	(17)
(18)		PrecStd	6.5	1.3	1.5	1.8	1.5	1.6	1.1	1.1	1.0	1.3	1.6	1.4	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	106.1	103.6	96.2	86.2	74.6	66.4	62.9	69.3	78.8	89.9	98.6	102.4	(19)	
(20)		MCWB	67.1	67.1	64.8	61.3	57.8	54.1	53.3	56.1	56.9	59.4	63.1	63.8	(20)	
(21)		DB	101.3	97.6	90.7	81.0	70.2	62.2	59.8	64.5	73.2	84.3	92.6	96.8	(21)	
(22)		MCWB	66.3	65.8	63.6	60.2	56.2	52.9	51.7	53.3	56.8	59.6	61.8	63.8	(22)	
(23)		DB	97.0	93.2	86.7	77.4	67.4	59.8	57.4	61.2	69.4	79.2	87.5	92.4	(23)	
(24)		MCWB	65.4	65.1	62.9	59.0	55.4	52.1	50.2	51.7	55.6	58.3	61.4	62.8	(24)	
(25)		DB	92.8	89.4	83.2	73.8	64.2	57.3	55.1	58.5	65.8	74.7	82.8	88.2	(25)	
(26)		MCWB	64.8	64.4	61.9	57.7	54.4	51.0	49.3	50.3	54.4	57.3	60.3	61.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	72.2	73.7	70.4	65.6	62.1	57.6	56.2	57.9	62.0	65.7	68.9	70.4	(27)	
(28)		MCDWB	88.4	81.4	80.6	76.3	65.5	60.9	59.3	65.1	70.9	76.3	82.4	81.6	(28)	
(29)		WB	70.5	70.9	68.0	63.0	59.2	55.2	53.3	55.1	59.8	63.2	66.8	68.2	(29)	
(30)		MCDWB	86.8	83.9	78.6	72.8	66.0	59.5	57.9	62.1	68.4	74.1	79.8	81.1	(30)	
(31)		WB	69.1	69.1	66.2	61.4	57.1	53.4	51.4	52.8	57.6	61.2	64.7	66.5	(31)	
(32)		MCDWB	86.1	82.9	78.9	71.8	64.8	58.2	55.8	59.8	65.9	72.5	77.2	81.7	(32)	
(33)		WB	67.7	67.5	64.6	59.6	55.3	51.7	49.8	50.9	55.3	59.3	62.9	64.7	(33)	
(34)		MCDWB	85.9	82.0	77.8	71.1	62.5	56.3	53.9	57.3	63.6	70.9	76.9	82.1	(34)	
(35)	Mean Daily Temperature Range	MDBR	26.1	24.2	24.3	23.5	20.5	17.1	16.7	19.5	22.6	25.4	25.5	26.7	(35)	
(36)		5% DB	30.6	28.4	28.1	27.1	23.8	20.9	20.1	24.1	28.1	32.1	31.8	30.9	(36)	
(37)		MCWBR	9.2	8.9	10.0	11.3	12.1	12.6	12.7	14.2	14.5	12.9	10.5	9.8	(37)	
(38)		5% WB	24.4	21.4	21.2	21.4	18.8	16.5	16.3	21.0	22.0	23.7	23.4	23.8	(38)	
(39)		MCWBR	9.3	8.4	9.4	9.9	10.5	10.8	11.0	13.1	12.9	11.9	9.6	9.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.359	0.344	0.323	0.309	0.299	0.289	0.285	0.295	0.320	0.327	0.345	0.343	(40)	
(41)		taud	2.433	2.512	2.550	2.552	2.542	2.586	2.597	2.541	2.459	2.463	2.455	2.476	(41)	
(42)		Ebn at Noon	310	308	303	289	274	269	278	290	298	310	312	316	(42)	
(43)		Edh at Noon	39	34	31	27	24	21	22	27	33	36	37	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2437	2124	1796	1331	940	712	778	1066	1515	1956	2222	2456	(44)	
(45)		RadStd	131	139	113	67	51	69	44	90	141	160	147	96	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.88	N/A	N/A	+2.30	+0.70	+0.92	N/A	N/A	N/A	N/A	(46)
(47)	Regional (22 neighbors)	+0.72	N/A	N/A	+1.89	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page